Confinement scaling as a linear algebra problem: the model that wins on cross-validation loses on a new machine

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Abstract

Energy confinement time sets the size a tokamak must reach to ignite, and the field predicts it with power-law scaling laws fitted across existing devices. On the ITPA global H-mode database (HDB5, standard analysis set STD5; 6228 quasi-stationary slices from 4471 discharges across 18 tokamaks) a random forest beats the published IPB98(y,2) scaling law by 41% in RMSLE under grouped cross-validation by discharge. We show this margin does not survive the question a scaling law exists to answer. Under leave-one-tokamak-out the ordering of the three fitted models is exactly reversed: the random forest is worse than a log-linear power law on 13 of 13 held-out machines, with a paired 95% interval of $[+0.157, +0.342]$ on the gap. We characterise the failure three ways. Its errors track the Mahalanobis distance of the held-out machine ($\rho = +0.85$) where the power law's do not ($\rho = -0.06$); a tree ensemble is bounded by its training target range, and on the largest device 48% of rows lie above anything it can output; and a polynomial ladder shows what flexibility costs is the tail of the error distribution rather than its centre. Ordering the machines by size reproduces, inside the database, a size extrapolation matching the 1.82× jump from the database to ITER; at that cut the tree ensembles score closer to a constant predictor than to the power law. We then build the model the diagnosis implies: a power law plus a bounded, heavily damped correction on its log residuals. At the ITER-matched cut it scores 0.206 against plain ridge's 0.278, the best blind model there, and we show the mechanism is the same boundedness that disqualifies trees as predictors, now applied to a residual centred on zero. Finally, split-conformal intervals calibrated to 90% under cross-validation cover 35% of rows on an unseen machine and 3% across the ITER-matched cut, at essentially unchanged width: the models are not vague out of distribution, they are confidently wrong. We then show that one line of dimensional analysis outperforms every model built here: imposing the Connor–Taylor constraints, which a scaling law expressible in dimensionless variables must satisfy, as linear equality constraints on the exponents gives 0.183 at the ITER-matched cut, the best blind score in this study, with no hyperparameter to tune. Calibrating the intervals on held-out machines repairs coverage to within two points of nominal on an unseen machine and, as the diagnosis predicts, fails to repair it across the size cut. The reversal reproduces on 5358 rows the standard set does not contain and again in a different confinement regime against a different published law. Finally we record a locked prediction for SPARC, JT-60SA and ITER: inside the database's size range the models agree to within 15%, and on ITER they disagree by a factor of 8.3. Separately, we report that the ten-feature design matrix used here is rank deficient by exactly two, one dependency being that a published scaling law added as a model feature is a fixed linear combination of the features already present.

1 Introduction

A confinement scaling law is a power law in the engineering parameters,

$$\tau_E = C\, I_p^{a_1} B_t^{a_2} \bar{n}_e^{a_3} P^{a_4} R^{a_5} \epsilon^{a_6} \kappa^{a_7} M^{a_8}, \tag{1}$$

so in log space it is exactly a linear model and fitting one is ordinary least squares on a log design matrix. The reference is IPB98(y,2) [1], which was used to size ITER.

Replacing Eq. (1) with a flexible regressor is an obvious move and, measured conventionally, it works. It is also the wrong measurement. Cross validation on this database holds out *discharges*, and every machine in a held-out fold also appears in the training fold, so it quantifies how much of JET is predictable from the rest of JET. A scaling law is not for the machines that exist. This paper escalates what is held out (a discharge, then a whole machine, then an entire size range) and reports what each escalation costs.

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2 Data and method

Data. The ITPA Global H-mode Confinement Database, standard analysis set STD5 v5.2.3 [2], obtained from OSF. After requiring finiteness and positivity, 6228 rows from 4471 discharges across 18 tokamaks remain. The target is the thermal confinement time `TAUTH`. No synthetic data appears in any result. The analysed file is pinned by SHA-256 and verified on load, so every number below refers to one specific set of bytes rather than to whatever the host currently serves.

Features and models. Nine log engineering features: plasma current, toroidal field, line-averaged density, loss power, major radius, elongation, inverse aspect ratio, effective mass and minor radius. The analytic IPB98 prior is *excluded* as a feature throughout the extrapolation arms, because its exponents were fitted on this same database including whichever machine is held out; retaining it leaks the answer into every fold. The zoo is a ridge regression on the log features (i.e. a fitted power law), a random forest (300 trees), a histogram gradient booster, polynomial ridge controls of degree 2 and 3, and a mean baseline. IPB98(y,2) evaluated analytically is reported as a physics reference, and is marked throughout as *not blind*.

Splits. (i) Grouped cross-validation by discharge. (ii) Leave-one-tokamak-out over the 13 machines with at least 30 rows. (iii) A size-ordered cut: machines are ordered by median major radius, and the model is trained below a cut and scored on every machine above it. Scores are RMSLE, $\sqrt{\overline{(\log\hat{\tau} - \log\tau)^2}}$. Intervals are percentile bootstraps resampling whole *machines*, since the claim concerns behaviour on an unseen machine.

3 The design matrix is rank deficient by two

Standardised, the ten-feature matrix has rank 8, with the last two singular values at 4.9×10^{-14} and 2.4×10^{-14}. Projecting candidate vectors onto the null space confirms two exact dependencies at relative residuals of order 10^{-16}: minor radius is derived as $a = \epsilon R$ during cleaning, and the IPB98 prior is a fixed log-linear combination of the other eight features.

The second is the substantive one. Adding a published physics scaling as a model feature feels like adding knowledge; in log space it provably is not, and it contributes exactly nothing to a log-linear fit. Nothing failed loudly because `scipy.linalg.lstsq` inverts through the SVD pseudoinverse and silently returns the minimum-norm member of a two-parameter family of coefficient vectors that fit identically well.

Two methodological traps are worth recording because both were fallen into first. The null space is a *subspace*, so when the deficiency exceeds one the basis vectors printed by the SVD generally do not resemble the dependency sought even when it is exactly right; the correct test is $\|v - N^{\top}Nv\|$. And standardisation rescales the null space, so a raw dependency must be mapped into standardised coordinates before comparison. Omitting this gives residuals of 0.41 and 0.96 for dependencies that are exactly true.

4 Refitting, and where the disagreement lives

Refitting Eq. (1) on all STD5 rows (design matrix 6228×9, condition number 10.7) gives I_p 1.08 against the published 0.93 and R 1.58 against 1.97, while P and B_t return essentially exactly. This disagreement is expected: IPB98 was fitted to a selected ITER-relevant subset, whereas this fit applies no selection criteria and includes the spherical tokamaks. A refit on a different population is a different regression.

The normal equations, QR and the SVD pseudoinverse were each implemented from scratch, including the triangular substitutions, and agree to 7.6×10^{-13}. Cholesky on the normal equations is five times faster and three orders of magnitude less accurate, which is the expected trade: it forms $X^{\top}X$ and so works at the *square* of the condition number. At $\kappa = 10.7$ that costs nothing real, and it is the only one of the three that fails loudly on a rank-deficient matrix.

The informative question is *where* the two differ. Decomposing the difference between the exponent vectors along the singular directions of the design matrix, 77% of it lies in the single weakest direction, which carries 0.3% of the matrix's variance, while the three strongest directions carry 82% of the variance and account for 0.75% of the disagreement. That weak direction is plasma current traded against machine size, which is weak for a structural reason: tokamaks are not designed to vary the two independently, so no number of additional shots from existing machines separates them.

Table 1: The same models and the same nine blind features under two splits. Only the held-out unit changes. IPB98 is a reference, not a blind competitor.

Model	CV	LOMO	Ratio	CV rank	LOMO rank
random forest	0.128	0.465	3.6×	1	5
hist gradient boosting	0.130	0.359	2.8×	2	4
ridge, log-cubic	0.148	0.849	5.7×	3	6
ridge, log-quadratic	0.158	0.300	1.9×	4	3
ridge, log-linear	0.181	0.214	1.2×	5	2
IPB98(y,2), analytic	0.199	0.188	1.0×	6	1
mean baseline	0.869	0.994	1.1×	7	7

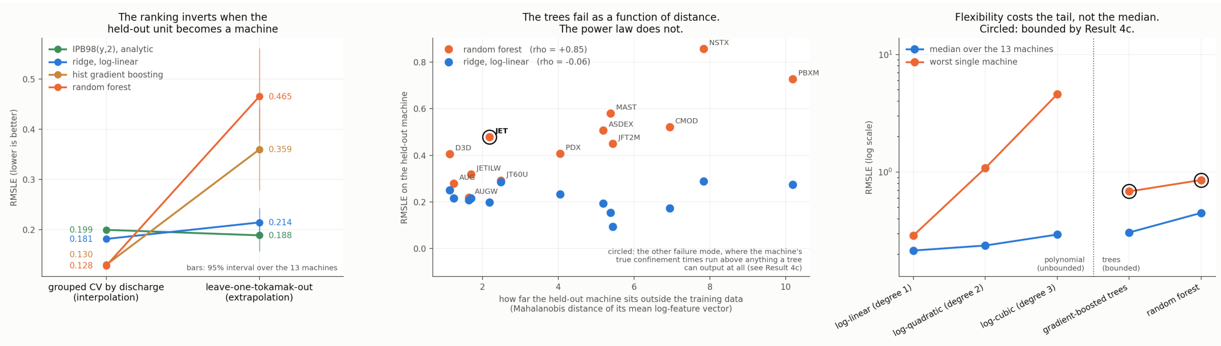

Figure 1: Interpolation against extrapolation. Left: each model under both splits; the lines crossing is the result. Middle: per-machine error against Mahalanobis distance from the training data, rising for the trees and flat for the power law. Right: the flexibility ladder, where what grows with flexibility is the tail rather than the centre.

5 The ranking inversion

Table 1 is the central result. Among the three models that actually fit something and are genuinely blind, the order under one split is the exact reverse of the order under the other. The random forest is the best model in the study by cross-validation and the worst of the three on a machine it has not seen.

Because the machines differ enormously in difficulty, the marginal bootstrap intervals overlap and are the wrong test; resampling the *paired* per-machine difference removes that shared difficulty. The random forest is worse than the log-linear power law on **13 of 13** machines, mean gap +0.251, 95% interval $[+0.157, +0.342]$. The gradient booster is worse on 12 of 13.

5.1 Three mechanisms

Distance. Ranking each model's per-machine error against the Mahalanobis distance of that machine's mean log-feature vector from the training mean gives $\rho = +0.85$ for the random forest and $+0.54$ for the gradient booster, against $\rho = -0.06$ for the log-linear power law. The trees' errors are explained by extrapolation distance. The power law's are not: its error is uncorrelated with how far the machine sits from anything it saw.

On the four most distant machines (MAST, C-Mod, NSTX, PBX-M, all small or spherical) the forest's error is 2.6 to 3.8× the power law's; on the four nearest it is within 1.6×.

A hard bound. A tree ensemble predicts by averaging training targets, so every prediction it can make lies in $[\min y_{\text{train}}, \max y_{\text{train}}]$ whatever the features say. With JET held out, 48% of its rows exceed the maximum confinement time in the remaining 12 machines and its best shot is 3.7× above anything any tree in the forest can output. This is not a shortfall more data would close; it is the functional form.

Flexibility costs the tail, not the centre. A polynomial ladder in the log features (degrees 1–3) is flexible but still extrapolates. Degree 2 is *not* meaningfully worse than degree 1 on a typical machine: median 0.238 against 0.216, better on 8 of 13, paired interval $[-0.013, +0.237]$ containing zero. What flexibility costs is the tail. Degree 1's worst machine is 0.289; degree 2's is 1.083 and degree 3's is 4.601, both on C-Mod. Boundedness is simultaneously why trees cannot extrapolate and why they cannot blow up: their worst cases, 0.686 and 0.857, sit far below degree 3's.

Table 2: Per-machine RMSLE under leave-one-tokamak-out, ordered by Mahalanobis distance of the held-out machine's mean log-feature vector from the training mean (computed through the pseudo-inverse, since that covariance is singular by Sec. 3). The trees degrade with distance; the power law does not.

Held out	rows	dist.	IPB98	ridge	hist GB	forest
D3D	388	1.1	0.252	0.251	0.246	0.406
AUG	1377	1.2	0.197	0.216	0.281	0.279
AUGW	767	1.6	0.218	0.207	0.212	0.219
JETILW	866	1.7	0.257	0.216	0.244	0.318
JET	1762	2.2	0.148	0.198	0.487	0.478
JT60U	100	2.5	0.275	0.285	0.307	0.291
PDX	97	4.0	0.227	0.233	0.323	0.407
ASDEX	431	5.2	0.131	0.194	0.207	0.507
MAST	39	5.4	0.136	0.154	0.317	0.581
JFT2M	69	5.4	0.095	0.094	0.234	0.450
CMOD	45	6.9	0.119	0.173	0.569	0.521
NSTX	185	7.8	0.222	0.289	0.686	0.857
PBXM	59	10.2	0.172	0.274	0.559	0.727

Table 3: Three questions of increasing difficulty. *Skill* places each model between the mean baseline (0.00) and the analytic power law (1.00).

Model	shot	machine	larger machine	skill
IPB98(y,2)	0.199	0.188	**0.194**	1.00
ridge, log-linear	0.181	0.214	**0.278**	0.93
random forest	0.128	0.465	**0.938**	0.41
hist grad. boosting	0.130	0.359	**1.072**	0.31
mean baseline	0.869	0.994	1.459	0.00

6 The size extrapolation ITER asks for

Holding out JET still leaves twelve machines spanning much of its range, so leave-one-out extrapolates in identity while interpolating in size. ITER's major radius is 6.2 m against 3.40 m for the largest row here, a factor of 1.82. That factor is available inside the database: training on the 14 smallest machines (up to DIII-D, 1.865 m, 3498 rows) and predicting the 4 largest (up to JT-60U, 3.40 m, 2730 rows) demands a ratio of 1.823 against ITER's 1.824. The cut is selected by proximity to that ratio rather than by eye, so it is a property of the data.

The power law retains 93% of the distance from a constant predictor to the analytic law; the trees retain 41% and 31%. The gradient booster scores 1.072 where predicting a single constant scores 1.459. Asked the question a scaling law exists to answer, the best cross-validated models land closer to a constant than to the law they beat by 41%. The ordering is identical on all three individually scored held-out machines, so the pooled number is not an artifact of JET's weight, and dropping the spherical tokamaks moves the random forest from 0.938 to 0.936, so the effect is size rather than plasma shape.

7 A power law with a bounded correction

The diagnosis implies a cure. Fit the power law, learn a correction on its log residuals, and damp it by λ:

$$\log \tau = \underbrace{f_{\text{ridge}}(x)}_{\text{unbounded, log-linear}} + \lambda \underbrace{g(x)}_{\text{correction on residuals}} . \tag{2}$$

At $\lambda = 0$ this is exactly the ridge row of Table 1. Two correction families differ in precisely the property that matters: a degree-2 log polynomial (unbounded) and depth-2 boosted trees (bounded by the residual range they were trained on).

Across the ITER-matched cut the two go opposite ways. Both start at 0.278; the polynomial correction climbs to 0.356 and the bounded correction falls monotonically to **0.206**, better than plain ridge on all three held-out machines and by the widest margin on JT-60U, the most distant (0.440 to 0.281). It is the best blind model at that cut, 26% below plain ridge and 4.6× below the random forest, and within 6% of IPB98, which was fitted with those machines included.

Table 4: Empirical coverage of nominal 90% split-conformal intervals, and the multiplicative half-width under leave-one-out.

Model	CV	LOMO	ITER cut	width
IPB98(y,2)	90%	89%	88%	1.40×
ridge, log-linear	90%	83%	70%	1.33×
bounded hybrid	90%	64%	76%	1.26×
hist grad. boosting	91%	45%	**0%**	1.23×
random forest	91%	35%	**3%**	1.23×

The mechanism is measurable. Fitted on the 14 small machines, the base power law is *biased* on the held-out ones: mean log residual -0.218, so it over-predicts by about 20%, while its scatter (0.172) is no worse than in-sample (0.187). The tree correction never leaves the range it was trained on, on any held-out row, and inside that bound it points the right way and is largest exactly where the bias is largest. **The same boundedness that makes a tree ensemble useless as a predictor of τ on a larger machine makes it safe as a corrector**, because the quantity it is bounded on is no longer a target that grows with machine size but a residual centred on zero.

Two honest limits. Cross-validation selects $\lambda = 1$ in both families, while the rung best on a held-out machine is $\lambda = 0$; nothing in the CV signal points at it. And scored at every rung of the size sweep rather than only the matched one, the bounded hybrid wins at 5 of 8 well-powered cuts, not all, with no rule over those eight points separating wins from losses. The claim that survives is narrow: at the cut matching the jump to ITER, and robustly across all nine correction settings tried there, a bounded correction beats the power law and the reason is understood.

8 Calibrated intervals and their collapse

For a next-step device the point error is not the deliverable; the interval is. We calibrate split-conformal intervals on log residuals, holding back 25% of *discharges* to calibrate, which gives finite-sample coverage under exchangeability of the calibration and test rows.

That proviso is the result. Exchangeability holds under grouped CV, and every model lands within a point of nominal there, and that control is what licenses reading the rest of Table 4. It fails by construction when the test rows come from a machine the model never saw. The random forest's 90% interval then covers 35% of rows on an unseen machine and 3% across the ITER-matched cut; the gradient booster covers none of the 2730 rows at all.

Crucially the widths barely move: no model's interval changes width by more than 1.5% between the two arms, because the half-width is set by calibration rows drawn the same way in both. The intervals do not become vague out of distribution. They stay the same size and miss.

Coverage collapses along the same axis the errors grow, but only for the trees: $\rho = -0.77$ for the random forest and -0.54 for the gradient booster against $+0.27$ for the power law, mirroring the point-error correlations of Sec. 5.1 with the sign flipped. The power law's intervals are somewhat too narrow everywhere rather than catastrophically too narrow far away, and for a next-step device that distinction is most of what matters, because distance is the one thing known in advance. None of this is a defect in conformal prediction; it is an assumption being false, measured.

9 One line of dimensional analysis

Every model to this point learns its functional form from the data. None is told any physics. That is worth revisiting, because the field has known since Connor and Taylor [4] that a confinement scaling law is not free: requiring it to be expressible in the dimensionless plasma variables imposes *linear equality constraints on the exponents.* A physics assumption is therefore a matrix C, and the fit becomes

$$\min_b \|Xb - y\|^2 \quad \text{subject to} \quad Cb = d, \tag{3}$$

which the KKT system solves in closed form. Nothing new is needed to run the experiment: the solver is the one written by hand for Sec. 4.

We derive the constraints in code from the definitions of ρ^*, β and ν^* rather than transcribing exponents from a paper, which makes the derivation checkable against something external. IPB98(y,2) was published

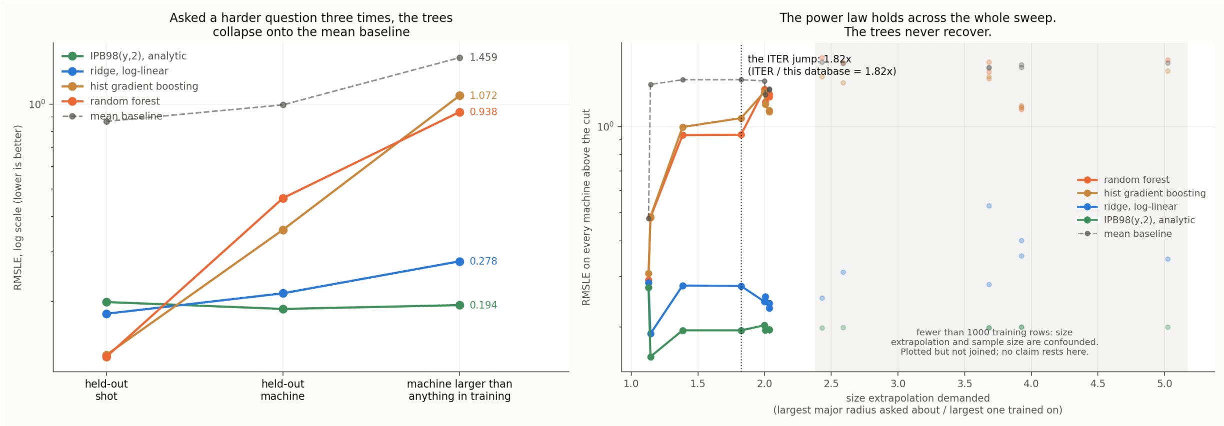

Figure 2: Size-ordered extrapolation. Left: the escalation across three splits. Middle: the per-machine breakdown at the ITER-matched cut. Right: the sweep across every cut, with underpowered cuts marked and excluded from claims.

Table 5: Fitting under the Connor–Taylor constraint hierarchy. The only difference between the first and last rows is the constraint: same features, same solver, no hyperparameter.

Model	in sample	CV	LOMO	ITER cut
power law, collisionless	0.1818	0.182	0.206	**0.183**
IPB98(y,2), analytic*	—	0.199	0.188	0.194
bounded hybrid (Sec. 7)	—	0.151	0.246	0.206
power law, electrostatic	0.2245	0.225	0.241	0.237
power law, Kadomtsev	0.1808	0.181	0.210	0.254
power law, unconstrained	0.1808	0.181	0.214	0.279

*fitted on this database including the held-out machines; not a blind model.

in 1999 and was never told about these surfaces. It lands on the Kadomtsev surface at a distance of 0.00096 and on the collisionless surface at 0.0045, both inside the rounding of its own two-decimal exponents. A transcription error would not reproduce that.

Table 5 is the result. At the ITER-matched cut the collisionless power law scores **0.183**, the best of any blind model in this study. It beats the bounded hybrid of Sec. 7, which was built specifically for that cut, and it beats the analytic law that was fitted with those machines included. It has no hyperparameter and nothing to tune.

The two constraints behave differently, and the difference is informative. The Kadomtsev constraint is *free*: at 0.1808 in sample it is identical to the unconstrained fit to four decimals, because the data already satisfies it unaided. It nonetheless beats the unconstrained fit at 15 of 15 cuts in the size sweep, since a constraint the data satisfies on average still stops the fit wandering when the training half is small. The collisionless constraint costs 0.001 in sample and wins at 8 of 8 well-powered cuts. Push one rung further to a beta-independent law and it degrades sharply, to 0.237. There is an optimum in the middle of the hierarchy, and nothing in this analysis predicted where it would fall; as with the hybrid, cross-validation does not select it, because every constrained fit is slightly worse than unconstrained under CV by construction.

It matters that this is a constraint and not a prior. Supplying the same physics as a Gaussian prior instead, shrunk along the weak singular direction of Sec. 4, is worth much less. Targeting the shrinkage beats isotropic shrinkage at every matched penalty strength, which is a real effect, but nothing in that family beats simply adopting IPB98's published exponents outright. A constraint names a surface; a prior names a point. The weaker information is worth more, because the surface is what the physics actually asserts.

10 Repairing the intervals, and the limit of the repair

The diagnosis in Sec. 8 makes a falsifiable prediction. If the interval collapse is really about exchangeability rather than about the models, then calibrating on held-out *machines* instead of held-out discharges should repair it, and should repair it only as far as that unit reaches. We test both halves.

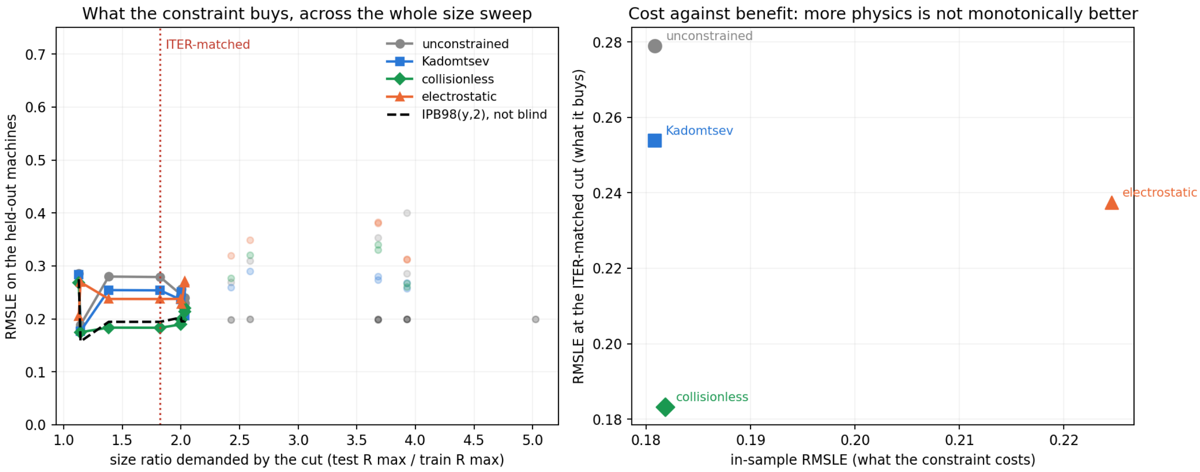

Figure 3: Cost against benefit across the size sweep. Left: the in-sample penalty each constraint imposes. Middle and right: the score at every size cut for the constrained and unconstrained fits, with underpowered cuts marked. The constraint costs almost nothing where the data is plentiful and pays where it is not.

Table 6: Empirical coverage of nominal 90% intervals under three calibration schemes, on a held-out machine and across the ITER-matched size cut.

	split conformal		by machine		+ distance	
Model	LOMO	cut	LOMO	cut	LOMO	cut
random forest	35%	3%	**88%**	29%	88%	40%
hist grad. boosting	45%	0%	**90%**	4%	89%	13%
ridge, log-linear	83%	70%	91%	77%	92%	82%
power law, collisionless	85%	91%	91%	**95%**	92%	92%
IPB98(y,2)*	89%	88%	89%	89%	89%	87%

Both halves land. On a held-out machine the repair is essentially complete: every model returns to within two points of nominal, tree ensembles included, and the random forest goes from 35% to 88%. Across the size cut none does. No recalibration makes the two exchangeable there, because every calibration machine is smaller than every test machine, and scaling the interval by extrapolation distance recovers only part of the gap (3% to 40% for the forest). A repair that stops exactly where the diagnosis says it should is stronger evidence for the diagnosis than one that worked everywhere would be.

The constrained power law of Sec. 9 is the single exception in Table 6: its intervals hold at the ITER-matched cut under every scheme, including plain split conformal, which fails there for everything else.

11 Replication on rows the standard set does not contain

Everything above rests on one file, which is the honest ceiling on all of it. The same OSF project publishes the full database revision, DB5.2.3, with 14153 rows against STD5's 6228, and we pin it by its own SHA-256. Matching on (tokamak, shot, time) shows STD5 is an ELMy-H-mode quality selection out of it and leaves two populations this study had not analysed: **5358 H-mode rows STD5 does not contain** (12 machines, zero row overlap) and **3860 ohmic and L-mode rows** in a different confinement regime, which we score against ITER89-P [5] because an H-mode law is the wrong baseline for an L-mode plasma.

This replication has exactly one way to be silently fake, and it is worth naming. If the row match against STD5 fails, every STD5 row stays in the "disjoint" arm, the arm becomes a superset of the data Sec. 5 was computed on, and it reproduces Sec. 5 by construction. We therefore check the matcher positively as well as negatively: STD5 matched against itself must overlap completely, and the disjoint arm must share zero rows.

The ranking inverts in both arms. On the disjoint H-mode rows the best cross-validated model beats the published law by 42%, within a point of the 41% this paper opens with, computed on rows that headline never saw, and the best model on an unseen machine is IPB98(y,2). On the ohmic and L-mode rows the cross-validation gain over ITER89-P is 67% and the best model on an unseen machine is the collisionless

Table 7: Predicted energy confinement time, with nominal 90% intervals. Locked 31 August 2026 against a fixed dataset digest.

Model	SPARC 1.85 m	JT-60SA 2.96 m (operating)	ITER 6.2 m
IPB98(y,2)*	0.765 s	0.479 s	**3.591 s**
power law, collisionless	0.724 s	0.428 s	2.837 s
ridge, log-linear	0.720 s	0.444 s	2.858 s
random forest	0.136 s	0.449 s	**0.435 s**
hist grad. boosting	0.305 s	0.418 s	0.444 s

power law of Sec. 9. Counting machine–model pairs where a tree ensemble loses to the published law on an unseen machine: 19 of 24 and 10 of 10.

So the reversal is not an artifact of the standard set's selection criteria, and not a property of ELMy H-mode or of IPB98(y,2) specifically. It is emphatically *not* an independent database: both arms come from the same ITPA collection and the same devices, and the five-machine arm is too small to carry a claim alone.

12 A locked prediction for three machines that have no data

Everything above is retrospective. To make the disagreement falsifiable rather than merely described, we record what each model predicts for three real devices, with intervals, a date, and a digest over the input rows so that a later edit leaves a mark. The parameter sets are published design values, and each reproduces the confinement time its own source quotes: SPARC to 0.6% and ITER to 2.9%.

Table 7 is the argument of this paper in one place. On JT-60SA, which sits inside the database's size range and is already operating, all five models agree to within 15%. On ITER, 1.82× beyond it, they disagree by a factor of **8.3**.

That gap cannot be closed by tuning, and Sec. 5 says why: a tree ensemble predicts by averaging training targets, so its output cannot exceed the largest one, which here is 1.321 s. The random forest that beats the published law by 41% under cross-validation is arithmetically incapable of returning ITER's predicted confinement time whatever it is asked, and its nominal 90% interval at ITER runs from 0.19 s to 0.98 s, containing neither the physics answer nor anything near it.

The JT-60SA column is the one that becomes checkable first, and it is worth being precise about when. As of September 2026 it is not checkable: the first JT-60SA campaign ran L-mode divertor plasmas, and its published power balance reports confinement following L-mode scaling, whereas the row above is an H-mode prediction scored against IPB98(y,2). H-mode operation is planned for the second campaign, beginning in the second half of 2026. The prediction fails if a stationary JT-60SA H-mode near this design point reports a thermal confinement time outside 0.349 s to 0.657 s. All five models place their intervals across roughly that range, so such a measurement would fail all of them together and one inside separates none of them: this row tests whether any model is right, not which, and that is why the ITER column carries the argument.

One feature of this table was not anticipated by anything above: **SPARC sits further from the training data than ITER does**, at a Mahalanobis distance of 6.9 against 4.7, despite being the smaller machine. Its 12.2 T field is far outside a database that tops out near 4 T. Size is not the only direction in which a next-step device leaves the data behind, and an extrapolation audit organised around size alone would have missed it.

13 Limitations

The refit population is not IPB98's, and the refit exponents should not be read as a correction to it. Two machines (JET and ASDEX Upgrade with their variants) supply 77% of rows. Only 13 of 18 machines are scored under leave-one-out; the five excluded are the smallest and most unlike the rest, so the reported gap is if anything optimistic. Thirteen machines is a small bootstrap and the paired differences are the more reliable statistic. The ITER-matched cut reproduces ITER's size ratio and nothing else: ITER is a burning plasma dominated by alpha self-heating at a gain no row here approaches, so clearing this bar is necessary and not sufficient. The per-machine coverages in Sec. 8 are proportions over as few as 39 rows and carry binomial uncertainty of ten points or more; the pooled contrasts are far larger than that noise, the within-column ordering is not. The constraint hierarchy of Sec. 9 has an optimum in its middle that nothing here predicted and that cross-validation cannot select, so the result is that a constraint helps rather than that this constraint is the right one. The replication of Sec. 11 is disjoint in rows but not independent in provenance: both arms come from the same ITPA collection and the same devices, and its five-machine arm is too small to carry a claim alone. The locked forecast of Sec. 12 is a record, not a validation; only JT-60SA can be checked against measurement in the near term, that check requires the H-mode campaign beginning in late 2026 rather than the completed L-mode one, and it is the one device of the three that sits inside the database's size range.

14 Conclusion

On real confinement data, the model that wins cross-validation by 41% loses to a 1998 power law on every machine it has not seen, and the failure has three measurable mechanisms rather than a tuning fix. At the size extrapolation that separates this database from ITER, the best cross-validated models land closer to a constant predictor than to the power law, and their nominal 90% intervals cover 3% and 0% of the held-out rows at unchanged width. A power law is not used in this field because it fits better, since it fits worse, but because it is the functional form that survives leaving the data behind. A bounded correction on its residuals improves it in the one direction a next-step device lies in, and the same boundedness that disqualifies trees as predictors is what makes that correction safe.

The strongest repair, though, is not a model at all. Constraining the exponents to a surface the physics requires them to lie on beats every model built here at the ITER-matched cut, costs 0.001 in sample, and has nothing to tune. Supplying the same physics as a prior is worth substantially less: a constraint names a surface where a prior names a point, and the weaker information is the more useful one. The reversal itself reproduces on rows the standard set does not contain and in a second confinement regime, so it is a property of how these models are validated rather than of one file. What the models disagree about is recorded rather than argued: on a machine inside the database's size range they agree to within 15%, and on ITER they differ by a factor of 8.3.

Data and code availability

The HDB5 STD5 dataset is third-party data obtained from OSF (`https://osf.io/drwcq`) and is not redistributed here; it is pinned by SHA-256 and verified on load. All code, the full writeup with every table and limitation, and the scripts that regenerate every number and figure in this paper are available in the accompanying repository.

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